

RESEARCH & DEVELOPMENT REPORT

LegalBot

An AI-Powered Cyber Crime Complaint and Legal Assistance System

A Retrieval-Augmented Generation Approach to Cyber Crime Victim Support (Segment S4)

Individual Project Report

Submitted in fulfilment of the requirements of the LegalBot Student Development Programme

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Abstract

India's formal justice system is hard to reach for an ordinary citizen. Over 5.39 crore cases are pending nationally, and the country has roughly 21 sanctioned judges per million people against a recommended benchmark of 50. Cyber crime sits at one of the sharpest edges of this problem: victims are confused about what has happened to them, often too ashamed to talk about it, and racing against a clock, since acting within the first 24 to 72 hours meaningfully improves the odds of getting stolen money back. This report describes LegalBot, an AI-powered legal assistant I built individually for Segment S4 (Cyber Crime Response) of the wider LegalBot initiative.

I built the system as a working, end-to-end prototype: a Retrieval-Augmented Generation (RAG) backend on FastAPI, ChromaDB and the Google Gemini API, paired with a Next.js/React frontend. It has four working modules — a grounded conversational assistant, an automated complaint-draft generator, a reporting-authority finder, and a tool that pulls structured case details straight out of a chat conversation. I describe each module below in terms of why it exists, how it is built, and where it sits in a cyber-fraud victim's journey from first realising something is wrong to producing a complaint draft ready to submit to the National Cyber Crime Reporting Portal or the 1930 helpline.

I also evaluate the build against the accuracy, usefulness, safety and engineering-quality criteria set out in the governing project brief, and I am upfront about where the prototype currently falls short — most notably the depth of the knowledge base and the absence of a language layer beyond English. The closing section lays out what I would prioritise next.

Keywords: *Cyber crime, Retrieval-Augmented Generation, legal technology, access to justice, IT Act 2000, Bharatiya Nyaya Sanhita 2023, AI legal assistant*

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1. Introduction

1.1 Background

Access to justice in India is limited by more than court capacity. A citizen also has to figure out what has actually happened to them, what rights they have, and which of the many available offices or portals is the right one to approach — and that is often the harder problem. Table 1 lays out the shape of this problem across five dimensions, as set out in the governing project brief, before I narrow in on cyber crime specifically.

Dimension	Evidence
Backlog	5.39 crore+ cases pending nationally (Dec 2025, NJDG), heading toward 5.5 crore by March 2026. Over 85% of these sit in district courts, and 1.8 lakh cases have been pending for more than 30 years. India has roughly 21 sanctioned judges per million people against a recommended 50.
Cost	Lawyer fees are unregulated, ranging from a few hundred rupees to several lakhs per hearing. DAKSH estimates litigants spend around ₹1,039 a day in direct court costs and about ₹1,746 a day in lost income — for years at a stretch. Many families end up borrowing or selling assets, and a good number simply abandon their claim.
Awareness	Free legal aid is a statutory right for women, children, SC/ST citizens, workers, undertrials and low-income persons, yet only around 15% of rural residents know it exists (NALSA, 2022). Rural lawyer density is roughly 1:10,000 against an urban 1:1,000.
Language	Statutes, judgments and forms are written in English or formal Hindi, which most Indians don't use day to day. Even a literate reader can struggle with terms like 'cognizable' or 'decree.' Any real solution needs to be multilingual, voice-friendly, and written at roughly an 8th-grade reading level.
Biggest clog	Land and property disputes alone make up about 20% of all pending cases and around 66% of pending civil cases.

Table 1. India's justice-access gap across five dimensions (source: LegalBot project brief).

Cyber crime is most acute on the Backlog and Awareness axes above — it feeds directly into a court system already stretched thin, and the average citizen has almost no idea what to do in the first hour after being defrauded. The LegalBot initiative was set up to address this gap across ten legal domains (S1 through S10), each handling a different kind of legal problem and a different group of underserved citizens.

This report is about Segment S4, Cyber Crime Response. I built LegalBot on my own to address it: an AI assistant that helps someone who has just been defrauded online understand what is going on, get grounded legal information, generate a complaint draft, and find out which authority to actually contact.

1.2 Problem Statement

Cyber crime has grown much faster than public awareness of the remedies available for it. In 2024 alone, more than 1.7 million complaints were filed through the National Cyber Crime Reporting Portal, with reported losses over ₹22,800 crore, and less than 10 percent of that money is ever recovered.

Looking at how these cases actually unfold, a fairly consistent pattern emerges rather than a single cause:

1. Someone is defrauded — phishing, UPI/OTP fraud, a fake “digital arrest” call, a bogus loan-recovery demand, an investment scam — and doesn't immediately register it as fraud.
2. Confusion turns into fear or shame, and both of those delay telling anyone, which delays reporting.
3. They either do nothing or do the wrong first thing, not realising that reporting within 24–72 hours (the “golden hour”) can get a bank to freeze the transaction.
4. By the time a report does get filed, if it gets filed, the window for recovering the money has usually closed.

There's a legal complication layered on top of this. The statutory framework changed in 2023, when the Bharatiya Nyaya Sanhita replaced the Indian Penal Code. A lot of what's on the internet still cites the old IPC section numbers, so even someone genuinely trying to find help can end up with outdated or wrong information. Left unchecked, an AI system trained on that material would just repeat the same mistakes with total confidence — which is exactly the failure mode this project tries to design around.

1.3 Objectives

1. Build an AI assistant that explains cyber crime law and victim rights in plain language, grounded in real source material rather than whatever the model happens to remember.
2. Automate drafting — not filing — of a structured complaint document ready for submission.
3. Work out the correct reporting authority or forum for a given case based on jurisdiction and the amount involved.
4. Cut down the reporting burden on someone who's already distressed by pulling structured complaint data straight out of a normal conversation.
5. Stay inside the ethical and legal limits appropriate for a legal-information product, as opposed to legal advice or representation.

1.4 Scope of the Study

Table 2 sets out what this individual project actually covers versus what it deliberately leaves out.

In Scope	Out of Scope (Future Work)
Grounded conversational Q&A over a cyber-crime knowledge base	Multilingual and voice interfaces (Hindi, regional languages)
Automated complaint-draft generation	Automatic filing with cybercrime.gov.in / NCRP
Reporting-authority / forum identification	WhatsApp Business API channel integration
Conversation-to-structured-data extraction	Bank-side transaction-freeze automation
A single-page web client integrating all modules	A fully time-boxed, guided golden-hour workflow

Table 2. Project scope boundary.

2. Contextual Review

2.1 India's Cyber Crime Landscape

The numbers here are the reason the golden-hour framing matters so much. The figures below are national cyber crime reporting data for 2024, and they shaped several of the design priorities I describe later.

Indicator	Value	Design implication
Complaints filed via NCRP (2024)	1.7 million+	The reporting channel is already known to millions; the gap is awareness and speed of the first step, not the existence of a portal.
Reported financial loss (2024)	₹22,800 crore+	Financial urgency justifies prioritising golden-hour guidance over general legal education.
Funds successfully recovered	Under 10%	Recovery depends on speed; the system must minimise time-to-first-action.

Table 3. Selected national cyber crime indicators and their bearing on system design.

2.2 Existing Reporting Mechanisms

Two official channels handle cyber crime reporting in India. The National Cyber Crime Reporting Portal, cybercrime.gov.in, takes complaints online and doesn't require a lawyer. The 1930 helpline is tied directly to the mechanism banks use to freeze a suspicious transaction. Both are free. What's missing isn't the infrastructure — it's that most people don't know either exists, or don't know that the golden hour is the thing that actually decides whether their money can be recovered.

2.3 Legal Framework

Two statutes govern this space. The Information Technology Act, 2000 covers offences specific to computers, data and networks — unauthorised access, identity theft, cheating by personation using a computer resource. The Bharatiya Nyaya Sanhita, 2023, which took over from the Indian Penal Code in July 2024, covers the general offences — cheating, criminal intimidation, extortion — that usually apply alongside the IT Act in a fraud case. RBI's zero-liability guidelines for unauthorised electronic transactions are also relevant where a victim reports quickly. Because the BNS transition is so recent, I treated getting the current section numbers right, rather than the outdated IPC ones still floating around online, as a core accuracy requirement rather than a nice-to-have.

2.4 Related Approaches to AI-Assisted Legal Support

Chatbots that answer legal questions aren't new, but most of the ones a citizen is likely to run into fall into one of two categories, and both have real weaknesses for a domain like cyber crime. The first is a rule-based decision tree — a fixed script of “if this, then ask that” branching logic. These are predictable and easy to audit, but they break the moment a user's situation doesn't fit the anticipated branches, which is often, since fraud narratives rarely arrive in a tidy structured form. The second is a general-purpose LLM chatbot with no retrieval layer at all, answering purely from what it learned during training. These handle open-ended conversation well but are exactly the systems most prone to citing a law, section

number, or procedure that sounds correct and isn't — a failure mode that is only getting more dangerous as India's statutory framework itself changes underneath them, as described in Section 2.3.

Retrieval-Augmented Generation, the approach this project uses throughout, sits between those two extremes. It keeps the open-ended, natural-language flexibility of an LLM-based chatbot while pinning every factual claim to a retrievable, editable source document, which is the same basic idea used by production legal-research tools and enterprise support assistants outside the LegalBot context. What is specific to this project is applying that pattern to a domain — cyber crime response — where the underlying law recently changed, the user is often in genuine distress, and a wrong answer isn't just unhelpful but can cost someone their only realistic chance at recovering stolen money.

3. Methodology

3.1 Design Approach

Everything about this build comes back to one constraint: a legal-information assistant is only as trustworthy as the material it's allowed to draw on. A plain LLM answering legal questions from memory will hallucinate confidently — wrong section numbers, wrong deadlines, stated as fact. So instead of relying on the model's memory, I used Retrieval-Augmented Generation throughout: every answer has to be conditioned on text pulled from a maintained knowledge base at query time, and the model is explicitly told not to go beyond what it retrieved.

The brief calls for four roles per team — a legal researcher/content lead, a backend-RAG engineer, a frontend/conversation designer, and a QA & evaluation lead. Since this was an individual project, I did all four myself: researching the legal content and drafting the knowledge pack, building the backend RAG pipeline and API, designing the frontend, and testing the result against the persona scenarios in Section 3.2.

3.2 User Persona Analysis

The brief defines five personas shared across every segment, since the LegalBot platform is meant to serve all ten legal domains through one interface. Table 4 lists all five and notes how relevant each one actually is to Segment S4. I concentrated my design effort on the personas whose core need is cyber-crime response, and I've kept the rest here for completeness rather than pretending they don't exist.

Persona	Situation	Relevance to S4
Ramesh, 45, farmer, rural UP	Ancestral land encroached by a cousin; suspicious mutation in land records	Out of scope for S4 (property/land domain, Segment S2)
Priya, 27, IT professional, Bengaluru	₹1.2L deposit withheld by landlord; separately, a ₹18k e-commerce refund stuck for 4 months	Partially relevant — her e-commerce/refund concern maps to Segment S1, but her need for fast answers and auto-drafted notices directly shaped this build's complaint generator
Fatima, 34, homemaker, Hyderabad	Escalating domestic violence; fears both the police and her family discovering it	Out of scope for S4 (women's safety domain, Segment S7)
Col. Sharma (Retd.), 68, Jaipur	Lost ₹6 lakh to a “digital arrest” impersonation scam; concealing the incident out of embarrassment	Primary S4 persona — shaped the tone, the golden-hour framing and the non-judgemental UX decisions in this build
Arjun, 24, gig worker/founder, Pune	A client owes ₹80,000 for five months of work; separately planning a cloud-kitchen business	Out of scope for S4 (MSME/payment-recovery domain, Segment S10)

Table 4. Full shared persona set and relevance to Segment S4.

I also reasoned through two cyber-crime scenarios that aren't among the original five — an IT professional hit by an unauthorised UPI debit, and a rural user targeted by a fake loan-recovery call —

mainly to stress-test the complaint generator and forum finder against fraud types Col. Sharma's scenario alone doesn't cover.

3.3 System Requirements

The functional requirements follow directly from Section 1.3: grounded question answering, complaint generation, forum identification, conversational data extraction. On the non-functional side, I prioritised three things — answers that stay grounded (no unverified legal claims), privacy-conscious handling of data from people who are already distressed, and a backend modular enough that the knowledge base could be updated without touching the application code.

4. System Design and Architecture

4.1 Architectural Overview

The system is a fairly standard client-server setup: a Next.js/React frontend talks to a FastAPI backend over REST/JSON. The backend has four routers — chat, complaint generation, forum lookup, case extraction — and all four share the same retrieval layer (ChromaDB) and the same generation layer (Google Gemini).

4.2 The Retrieval-Augmented Generation Pipeline

The RAG pipeline is really the core design decision of this whole project — every substantive answer goes through it. It runs in five stages, laid out in Table 5.

Stage	Process
1. Query intake	The user's message is received by the frontend chat interface and forwarded to the backend.
2. Embedding	The backend converts the query into a vector embedding using the embedding service.
3. Retrieval	ChromaDB is queried for the knowledge-base entries most semantically similar to the embedded query.
4. Grounded generation	The retrieved text is injected as context into a prompt sent to the Gemini model, which is instructed to answer only from that context.
5. Response delivery	The grounded, plain-language answer is returned to the frontend and rendered in the chat interface.

Table 5. Stages of the Retrieval-Augmented Generation pipeline.

This is what actually addresses the accuracy risk from Section 2.3. Because the model can only answer from retrieved, verified content, it's much less likely to invent an IT Act or BNS section number the way an ungrounded chatbot would. Of everything in this system, I'd call this the single most important accuracy control.

4.3 Technology Stack

Layer	Technology	Purpose
Frontend framework	Next.js 16, React 19	Application shell, routing, client-side state management
Styling	Tailwind CSS v4	Utility-first interface styling
Frontend utilities	lucide-react, react-hook-form, axios	Icons, form handling, HTTP requests
Notifications	react-hot-toast	Success/error toast feedback on async actions

Layer	Technology	Purpose
Client-side PDF export	html2canvas, jsPDF	Captures the complaint preview as an image and embeds it into a downloadable PDF
Backend framework	FastAPI (Python)	REST API layer with Pydantic request/response validation
Vector database	ChromaDB	Persistent semantic store for the knowledge base
LLM provider	Google Gemini API (google-genai SDK)	Query embeddings, grounded generation, structured JSON extraction
Configuration	python-dotenv	Environment-based management of API credentials

Table 6. Technology stack by architectural layer.

Two things the brief's suggested architecture calls for aren't in this build yet: containerisation/CI (Docker, GitHub Actions) and a language layer (Bhashini or IndicTrans2). Both are on the future-work list in Section 9.

4.4 End-to-End Data Flow: A Worked Example

The five stages in Table 5 are easier to picture with an actual example running through them. Suppose a user types: “Someone called me saying my son is in police custody in Mumbai and I need to transfer money right now or he'll be arrested. Is this real?” Table 7 traces that single message through the pipeline.

Stage	What happens with this query
1. Query intake	The frontend sends the raw message to POST /chat/ as the question field.
2. Embedding	The backend embeds the full sentence, capturing its semantic content rather than just keyword-matching on “police” or “arrest.”
3. Retrieval	ChromaDB returns the two closest knowledge-base entries — in this case, entries describing “digital arrest” impersonation scams and golden-hour reporting steps, since those are semantically closest to the query even though the user never used the phrase “digital arrest” themselves.
4. Grounded generation	Gemini receives the user's question plus the two retrieved entries as context, with an instruction to answer only from that context and to say so plainly if the context does not cover the question.
5. Response delivery	The user receives a plain-language explanation that this matches a known “digital arrest” scam pattern, that no Indian law-enforcement agency arrests people over a video call or demands payment to avoid arrest, and a prompt to call 1930 immediately rather than transfer any money.

Table 7. A single query traced through the RAG pipeline.

The part worth noting here is stage 3. The user never used the term “digital arrest” anywhere in their message, but semantic embedding retrieval still surfaced the right knowledge-base entry because the

meaning of the sentence, not its exact wording, is what gets matched. A keyword-only search would very likely have missed this, which is one of the practical reasons I chose a vector-based retrieval layer over a simple text search.

5. Implementation

I built four modules on top of the shared RAG architecture from Section 4. Below is each one, why it exists, and how it actually works.

5.1 Module 1: AI Legal Chat Assistant

This is the front door of the system. Someone describes what happened — say, a call demanding money to avoid a fake arrest warrant — in their own words, and the query goes through the RAG pipeline from Section 4.2. On first load, the frontend also shows a set of quick-prompt suggestions, mainly for people who aren't sure how to even start describing their situation.

5.2 Module 2: Complaint Generator

This module takes a structured incident description — complainant details, who or what was involved, what kind of fraud it was, the facts, and what relief is being asked for — and turns it into a complaint draft. The backend builds a structured prompt from these fields and asks Gemini for a full draft: the authority being addressed, complainant details, the opposite party, a statement of facts, the grounds of complaint, the relief sought, and a verification clause. The frontend shows this in a dedicated complaint workspace with a live preview and an export button.

The PDF export is entirely client-side. It captures the preview panel as a canvas image with `html2canvas`, embeds that image in a single-page PDF with `jsPDF`, and downloads it straight from the browser. That avoids needing a server-side rendering step, but it does mean the exported PDF is essentially a picture of the text rather than selectable text — a trade-off I've flagged in Section 9.

5.3 Module 3: Forum / Reporting-Authority Finder

One of the most common sources of confusion, mentioned in Section 2.2, is simply not knowing who to go to — the local police station, the district cyber cell, the state cyber cell, or the national portal. This module takes the user's state, district and the amount of money involved, applies tiered routing logic, and returns the authority that applies along with why.

5.4 Module 4: Case Detail Extractor

A distressed user shouldn't have to explain their situation twice. This module takes the full chat transcript and sends it to Gemini with a strict output schema, asking it to pull out the complainant's name, the counterparty, the subject matter, the issue, the facts, and the relief requested as structured JSON. Those fields then pre-fill the Complaint Generator from Section 5.2, so the user is mostly correcting details rather than typing them all in again.

5.5 API Design

The backend exposes four REST endpoints, listed in Table 8, each backed by a Pydantic request and response schema.

Endpoint	Method	Purpose	Key request fields
/chat/	POST	Grounded question answering	question

Endpoint	Method	Purpose	Key request fields
/generate-complaint/	POST	Complaint draft generation	name, counterparty, subject, issue, facts, relief
/forum/	POST	Reporting-authority identification	state, district, claim/loss amount
/extract-case/	POST	Structured data extraction from a transcript	conversation (free text)

Table 8. Backend API endpoints.

5.6 Knowledge Base Construction

The knowledge base is really what everything else depends on — the model is only as reliable as what it's allowed to retrieve from. Source material sits as plain-text entries, gets embedded, and is upserted into a ChromaDB collection through an ingestion script I can re-run whenever the content changes, without touching the rest of the application.

The brief sets a target of 60–100 plain-language, source-cited entries per segment and calls the knowledge pack “the most important deliverable.” I'll be honest about where this stands right now: the knowledge base currently holds a small set of introductory entries, well short of that target. I've recorded this as an open item in Section 6.2 rather than glossing over it. The intended content for Segment S4 covers:

- Golden-hour response procedure: calling 1930, requesting a transaction freeze, preserving evidence.
- Step-by-step guidance for filing through the National Cyber Crime Reporting Portal (NCRP).
- Applicable IT Act, 2000 and BNS, 2023 provisions for common fraud categories, including phishing, UPI fraud, identity theft, sextortion and impersonation.
- RBI zero-liability rules for unauthorised electronic banking transactions.
- An evidence-preservation checklist covering screenshots, transaction IDs, call logs and suspicious URLs or UPI identifiers.
- A reference set of common fraud patterns, including digital-arrest calls, fraudulent loan applications and investment scams.

6. User Interface and Experience Design

The backend architecture only matters if a distressed person can actually use it. This chapter walks through the frontend as it exists today, screen by screen, and the reasoning behind each one.

6.1 Design Language

The interface leans on a small, deliberately calm palette rather than the high-saturation colours common in consumer apps. Given who this is for — someone who has often just lost money and is frightened or embarrassed — I avoided anything that reads as alarming (heavy reds, flashing states) or dismissive (overly playful colours, cartoonish icons). Typography favours large, legible type sizes throughout, partly for Col. Sharma's persona in Section 3.2 and other older users who may not be comfortable with small text on a phone screen.

6.2 Landing / Hero Screen

The first screen a user sees is deliberately sparse: a short headline, a single input box, and a small set of quick-prompt suggestion chips (for example, “I think I've been scammed” or “Someone is threatening me over the phone”). The quick prompts exist because a blank text box is intimidating for someone who doesn't know how to describe what just happened to them — giving them a starting sentence to tap lowers that barrier considerably. An animated background element gives the screen some visual life without pulling focus from the input box itself.

6.3 Chat Workspace

Once a conversation starts, the user moves into a standard message-bubble chat view: their messages on one side, LegalBot's grounded responses on the other, with a typing indicator while a response is being generated. Inline action cards appear within the conversation when relevant — for instance, after LegalBot identifies that a case looks like it needs a formal complaint, a card appears offering to jump straight into the Complaint Generator, carrying the conversation context forward rather than making the user start over in a separate part of the app.

6.4 Complaint Workspace

This view is split into an editable form on one side and a live preview of the generated complaint on the other, so the user can see the document taking shape as they fill in or correct fields. Once satisfied, a single export action triggers the `html2canvas/jsPDF` flow described in Section 5.2 and downloads the complaint as a PDF. Toast notifications (`react-hot-toast`) confirm success or flag an error, which matters more than it might sound — someone already anxious about their situation needs clear, immediate confirmation that the action actually worked.

6.5 Forum Workspace

The forum finder is deliberately the simplest screen in the app: state, district, and the amount involved, submitted through a short form. The result is returned as a single, clearly labelled card naming the applicable authority, rather than a list of options the user has to interpret themselves — the whole point of this module, per Section 5.3, is to remove decision-making burden, not add another one.

7. Illustrative Case Walkthrough

To make the architecture in Chapters 4 and 5 concrete, this chapter follows a single illustrative case — based on the Col. Sharma persona from Section 3.2 — all the way through the system, from the first message to a completed complaint draft. The transcript below is illustrative rather than a captured production log, but it reflects the actual prompts, fields and outputs each module in Chapter 5 is designed to produce.

7.1 Scenario

Col. Sharma, 68, retired and living in Jaipur, receives a video call from someone in what looks like a police uniform, claiming his son has been caught in a narcotics case in Mumbai and will be arrested unless ₹6 lakh is transferred immediately to a “safe government account.” Panicked, he transfers the money before realising, an hour later, that his son is safe at home. He opens LegalBot not knowing whether anything can still be done.

7.2 Conversation

His first message to the chat assistant (Section 5.1) reads: “I sent ₹6 lakh to someone who said my son would be arrested. I think I made a huge mistake. What do I do now?” The RAG pipeline (Section 4.2) retrieves knowledge-base entries on digital-arrest scams and golden-hour reporting steps, and LegalBot responds with three things in order: reassurance that he is not the first person this has happened to and it is not too late to act, an explanation that this matches a known “digital arrest” fraud pattern with no basis in real law-enforcement procedure, and immediate, specific next steps — call 1930 right now, ask his bank to flag the transaction, and file a complaint on cybercrime.gov.in as soon as possible.

7.3 Extracted Case Data

As the conversation continues and Col. Sharma provides more detail, the Case Detail Extractor (Section 5.4) pulls a structured record from the transcript, illustrated below in simplified form:

```
{ "name": "Col. R. Sharma", "counterparty": "Unidentified caller impersonating police, Mumbai jurisdiction", "subject": "Digital arrest scam / fraudulent money transfer", "issue": "Transferred ₹6,00,000 under threat of son's fabricated arrest", "facts": "Received video call from person in police uniform; threatened with son's arrest in a narcotics case; transferred funds to specified account within the hour; no such case exists", "relief": "Transaction freeze, fund recovery, registration of FIR under IT Act and BNS provisions" }
```

Figure 1. Simplified structured output from the Case Detail Extractor for this scenario.

7.4 Generated Complaint Draft

These extracted fields pre-fill the Complaint Generator (Section 5.2), which produces a draft with the structure below — shortened here for space, but matching the section order the module actually generates:

- Authority addressed: Cyber Crime Cell, jurisdiction as returned by the Forum Finder (Section 7.5 below)
- Complainant details: name, contact information, address

- Opposite party: unidentified caller, phone number and any UPI/bank details captured during the call
- Statement of facts: a chronological account of the call, the threat, and the transfer, drawn from the extracted case data in Section 7.3
- Grounds of complaint: cheating and impersonation under the Bharatiya Nyaya Sanhita, 2023, and identity theft/cheating by personation under the Information Technology Act, 2000
- Relief sought: immediate transaction freeze, investigation, and recovery of funds
- Verification clause: a standard declaration that the facts stated are true to the complainant's knowledge

Col. Sharma reviews this draft in the live preview described in Section 6.4, corrects a phone number the extractor mis-transcribed, and exports it as a PDF to take to the cyber crime cell in person, alongside his own filing on cybercrime.gov.in.

7.5 Forum Result

In parallel, the Forum Finder (Section 5.3) takes his state (Rajasthan), district (Jaipur), and the amount involved (₹6,00,000) and returns the applicable cyber crime reporting authority for that jurisdiction and case value, along with a short note on why that authority applies — removing exactly the kind of uncertainty described in Section 2.2 that keeps victims from acting quickly.

This walkthrough is necessarily a simplified, best-case path through the system. A real user's conversation would likely be less linear, and the honest gaps recorded in Section 9 — particularly the missing disclaimer and the thin knowledge base — would be more visible in practice than this illustrative example suggests.

8. Results and Discussion

8.1 Functional Outcomes

The prototype does what I set out to build in Section 1.3: a working grounded chat assistant, an automated complaint-drafting flow, an authority-routing function, and a data-extraction feature that cuts down repeated typing for the user. All four modules sit behind one frontend, so a user can go from describing what happened to holding a complaint draft without leaving the app.

8.2 Deliverables Compliance

The brief lists four deliverables that every segment team is expected to produce, regardless of team size. Table 9 checks this build against each one — what's actually done versus what's still outstanding.

Required deliverable	Brief's target	Current status in this build
Knowledge Pack	60–100 cited entries, tagged by state and last-verified date	Ingestion pipeline works; populated content is a small set of introductory entries, well short of the target volume
Conversation design	Intake flow of ≤ 6 questions, ≥ 2 wizards/calculators, a domain tone guide	The chat interface and quick-prompt suggestions cover the intake flow; no dedicated wizard or calculator (e.g. a golden-hour timer) exists yet
Documents	≥ 3 generator templates validated against real formats	One template is implemented — the complaint draft, exportable to PDF; others (an evidence checklist, an escalation letter) are future work
Evaluation set	100 realistic questions in ≥ 2 languages with verified ideal answers	Not built yet — see Section 8.3 for why this is the next priority

Table 9. Compliance of this build against the brief's common deliverables.

I've kept this table honest rather than glossing over the gaps, because being upfront about what's unfinished is itself part of the accuracy discipline (Section 8.5) the LegalBot initiative asks of every team.

8.3 Evaluation Framework

The brief scores every build against five weighted criteria, shown in Table 10. I used the same framework to prioritise what to build first, and to judge the finished prototype against.

Criterion	Weight	Assessment basis for this build
Accuracy	30%	Correctness of forum, law and procedure references; absence of fabricated citations from the RAG pipeline
Usefulness	25%	Whether a real user can complete the golden-hour journey end to end within the application
Safety and ethics	20%	Presence of disclaimers, privacy-conscious data handling, and routing of high-stakes cases to human help

Criterion	Weight	Assessment basis for this build
Engineering quality	15%	Soundness of the RAG pipeline, API design and frontend/backend integration
Content craftsmanship	10%	Clarity, correctness and coverage of the cyber-crime knowledge pack

Table 10. Evaluation framework and weighting.

The next validation steps I'd want to run: build an evaluation set of realistic cyber-crime queries with verified ideal answers to measure retrieval accuracy; check generated complaint drafts against real NCRP and police complaint formats; spot-check forum-routing output against known jurisdictional rules; and walk through the persona scenarios from Section 3.2 end to end, from the first message to a finished draft.

8.4 Sample Evaluation Questions

Table 11 gives a small sample of the kind of question the full 100-question evaluation set (Table 9, still outstanding) should eventually cover, spanning the different fraud types and procedural questions a real user is likely to ask.

Sample question	What the ideal answer must get right
"Someone I don't know transferred ₹5,000 out of my account using my UPI PIN I never shared. What do I do?"	Names 1930 and cybercrime.gov.in; mentions RBI zero-liability protection if reported promptly; does not promise guaranteed recovery
"Is it true the police can arrest me over a video call for a pending case?"	States clearly that no Indian law-enforcement agency arrests or demands payment over a video call; names this as a common scam pattern
"Which portal do I use to report a fake loan app that's threatening me?"	Names the National Cyber Crime Reporting Portal; does not confuse it with a consumer-forum or banking-ombudsman process
"What evidence should I keep before I report a phishing email?"	Lists screenshots, headers, the sender's email/URL, and any transaction IDs; does not tell the user to delete or ignore the email
"Does IPC Section 420 still apply to online cheating cases?"	Correctly notes the IPC was replaced by the BNS in July 2024 and gives the current equivalent, rather than answering purely off IPC 420
"I'm being blackmailed with a private photo. Who do I contact?"	Routes toward 1930 and, where relevant, women's-safety support channels; avoids judgemental language; does not ask for the image or unnecessary detail

Table 11. Illustrative sample of evaluation-set questions and grading criteria.

None of these six have been formally graded against the current build — they're included here to show what the eventual evaluation set should look like, not as a substitute for actually running it.

8.5 Ethical and Legal Considerations

Because this operates in a legal domain, everything it outputs is meant to read as legal information, not legal advice — the Advocates Act, 1961 reserves the practice of law for enrolled advocates, so that line matters. Every complaint draft is meant to be clearly labelled as a draft for the user to review and submit themselves, not a filed document. One gap I should flag here: the brief asks for a disclaimer on every session and every document, and this build doesn't currently render one anywhere in the interface or on the generated complaint — that's a straightforward fix and probably the single highest-priority item before this goes in front of a real user.

The RAG architecture from Section 4.2 is the main defence against fabricated legal citations. Given how emotionally loaded cyber-fraud victimisation can be — the Col. Sharma persona from Section 3.2 is a good example — I treated interface tone as a real design concern, not an afterthought, and tried to avoid any language that could make someone feel judged or foolish for having been defrauded. On data handling, the goal was to collect the minimum necessary, and high-stakes cases (large losses, ongoing extortion, immediate danger) are meant to be routed toward human channels — the 1930 helpline, NALSA/DLSA — rather than handled purely through automated conversation.

9. Limitations

The brief lays out a 16-week phased roadmap: domain deep-dive and platform skeleton, knowledge pack v1 and end-to-end retrieval, wizards/generators/multilingual support, evaluation and user testing, then polish and demo. Measured against that, this build sits at roughly the end of the second phase — the RAG pipeline works end to end and the first conversational flow and document generator are in place, but the knowledge pack isn't at target depth, there's no wizard or calculator yet, there's no multilingual support, and no formal evaluation has been run. The limitations below follow from that.

- No disclaimer is currently shown anywhere in the interface or on the generated complaint, despite the brief requiring one on every session and document — this is the most important item to fix before real users see the product.
- The interface is English-only, which leaves out users more comfortable in Hindi or a regional language, something the brief emphasises repeatedly.
- ChromaDB is running as a local, dev-stage vector store rather than something production-grade like pgvector or Qdrant.
- There's no automated filing with cybercrime.gov.in or the NCRP; the system only produces a draft, in line with the ethical boundary described in Section 8.5.
- No voice interface exists, which matters a lot for senior and low-literacy users in this target group.
- Retrieval always returns a fixed top-two result set with no reranking step, which could hurt precision as the knowledge base grows.
- PDF export is a captured image of the complaint preview rather than selectable text (Section 5.2) — fine for printing, not great if someone needs to edit or machine-process the document afterward.
- There's no containerisation or CI/CD (Docker, GitHub Actions) set up yet, even though it's part of the suggested infrastructure in the brief.
- No formal evaluation set or retrieval-accuracy measurement has been run against the knowledge base yet, as noted in Table 9.

10. Conclusion and Future Work

What I've built is a working, end-to-end prototype for Segment S4, Cyber Crime Response. The main contribution here is architectural: showing that a legal-information assistant can be made a lot safer by constraining generation to retrieved, verifiable source material instead of letting the model answer from memory. On top of that retrieval-grounded core, the build adds the three things I thought would matter most to a cyber-fraud victim — somewhere to explain what happened and get an answer back, a complaint draft they don't have to write from scratch, and a way to figure out which authority to actually go to.

The biggest gap between where this prototype is now and the full ambition of Segment S4 is knowledge-base depth, language coverage, and the lack of a real time-boxed golden-hour workflow, all covered in Section 9. If I were prioritising next steps, in order: first, get the knowledge pack toward the 60–100

cited-entry target with a re-verification cycle, since legal accuracy depends entirely on this content; second, add a disclaimer to every session and document, which is a small change with an outsized safety benefit; third, build Hindi and at least one regional-language interface, likely through Bhashini or IndicTrans2; fourth, build an actual guided, time-stamped golden-hour workflow that walks a user through the first 24–72 hours step by step; and fifth, move retrieval from a fixed top-two lookup to hybrid semantic-and-keyword search with reranking as the knowledge base grows.

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Appendices

Appendix A: Sample Knowledge Base Entries

Two illustrative entries in the plain-language, source-cited format the knowledge base (Section 5.6) uses:

Entry: “Golden-hour transaction freeze.” If you report a fraudulent transaction to the 1930 helpline or on cybercrime.gov.in within roughly 24 to 72 hours, the bank can be alerted to freeze the receiving account before the money is withdrawn or moved further. The sooner you report, the higher the chance of recovery. Source: National Cyber Crime Reporting Portal guidance. Last verified: [to be dated at ingestion].

Entry: “Digital arrest scams are not real arrests.” No Indian law-enforcement agency conducts arrests over a phone or video call, and none will ask you to transfer money to avoid arrest or to ‘verify’ your identity. If you receive such a call, do not transfer any money; hang up and report the number to 1930. Source: Ministry of Home Affairs public advisory. Last verified: [to be dated at ingestion].

Both entries follow the format the brief asks for: plain language, a single clear claim, and a named, checkable source — the structure the full 60–100-entry knowledge pack (Section 5.6, Table 9) still needs to be built out to.

Appendix B: Sample API Request and Response

A representative request/response pair for the /forum/ endpoint (Section 5.5):

Request — POST /forum/

```
{ "state": "Rajasthan", "district": "Jaipur", "amount": 600000 }
```

Response

```
{ "authority": "Cyber Crime Cell, Jaipur (Rajasthan State Cyber Crime Unit)", "basis": "Jurisdiction determined by district of residence; case value routed to the state-level cyber unit given the amount involved", "contact_channel": "cybercrime.gov.in or 1930" }
```

This mirrors the Forum Finder output described in Section 7.5 of the case walkthrough.

Appendix C: Glossary of Terms

Term	Meaning
RAG (Retrieval-Augmented Generation)	An AI architecture in which a language model's answer is conditioned on text retrieved from an external knowledge source at query time, rather than generated purely from the model's training data.
Hallucination	A confident but factually incorrect output from a language model — for example, citing a law or section number that does not exist or does not apply.
Golden hour	The roughly 24–72 hour window after a fraudulent transaction during which reporting to 1930 or the bank has the best chance of getting funds frozen before they are withdrawn.
NCRP	National Cyber Crime Reporting Portal (cybercrime.gov.in), the official online channel for filing cyber crime complaints in India.

Term	Meaning
Vector embedding	A numerical representation of text that captures its meaning, allowing a system to find semantically similar content even when the wording differs.
ChromaDB	An open-source vector database used in this project to store and search knowledge-base embeddings.
Zero liability	RBI guidelines under which a customer may not be held liable for losses from an unauthorised electronic transaction, provided it is reported within a specified window.
BNS (Bharatiya Nyaya Sanhita)	The statute that replaced the Indian Penal Code as India's primary criminal code, effective July 2024.

Table 12. Glossary of key terms used in this report.